Chapter 9 Notes

Section 1

Notes

Here is why $A \cup B = A \cup (B - A)$.

On the one hand, suppose that $x \in A \cup B$. Then either $x \in A$ or $x \in B$. In the first case, obviously $x \in A \cup (B - A)$. In the second case, if $x \in A$, then we are done (by the first case). Otherwise $x \in B - A$, meaning $x \in A \cup (B - A)$. On the other hand, suppose that $x \in A \cup (B - A)$. Then obviously $x \in A \cup B$.

Section 2

Notes

Note that Lemma 2.2 assumes that h is defined on all subsets of A.

In proof 2 of the Bernstein-Schröder theorem, here is why $A - [A - g''(B - f''(X))] = g''(B - f''(X))$. Obviously, if $x \in A - [A - g''(B - f''(X))]$, then $x \in g''(B - f''(X))$. On the other hand, if $x \in g''(B - f''(x))$, note that $x \in A$, as g is a function from B into A. Thus $x \notin A - g''(B - f''(X))$, meaning $x \in A - [A - g''(B - f''(X))]$.

Exercise 2.1

(a) Consider any fixed point A of h. Then $h(A) = A$, meaning $A \subseteq h(A)$. Hence $A \in \Sigma$, and thus $A \subseteq B$. Thus B is the largest fixed point of h.

(b) Let Σ be the set of all subsets X of A such that $X \subseteq h(X)$ and that X is a subset of all fixed points of h. Let $B = \cup\Sigma$. We aim to show that B is a fixed point of h. Obviously, B is a subset of all fixed points of h (thus B is the smallest fixed point of h).

On the one hand, we show that $B \subseteq h(B)$ (this part is identical to the proof of the first part of the original lemma). So suppose that $b \in B$. Then $b \in X$ for some $X \in \Sigma$. Because $X \in \Sigma$, then $X \subseteq h(X)$, hence $b \in h(X)$. But $X \subseteq B$, and because h is monotone, it follows that $h(X) \subseteq h(B)$. Thus $x \in h(B)$. Hence $B \subseteq h(B)$.

On the other hand, we show that $h(B) \subseteq B$. Well, we have shown that $B \subseteq h(B)$, meaning $h(B) \subseteq h(h(B))$ (because h is monotone). Also, consider any fixed point A of h. Then $B \subseteq A$. Hence $h(B) \subseteq h(A)$. But $h(A) = A$, meaning $h(B) \subseteq A$. Thus $h(B)$ is a subset of every fixed point of h, and $h(B) \subseteq h(h(B))$, meaning $h(B) \in \Sigma$, and thus $h(B) \subseteq B$. Hence we have shown that $h(B) = B$, meaning B is a fixed point of h (and the smallest one).

Section 3

Lemma 0.1. For any classes A and B, $A \prec B$ and $B \prec A$ do not both hold.

Proof. This is immediate from the Bernstein-Schröder theorem. For suppose $A \prec B$ and $B \prec A$ both held. Then $A \preceq B$ and $B \preceq A$ both held, meaning, by the Bernstein-Schröder theorem, $A \cong B$. But this contradicts the fact that $A \cong B$ was assumed to not hold (as $A \prec B$ was assumed to hold). □

Lemma 0.2. For any class A, no subclass of A is greater than A.

Proof. Suppose a subclass B of A was greater than A. Then $A \prec B$ holds. Also $B \prec A$ holds (this is because $B \preceq A$ holds as $B \subseteq A$, but $B \cong A$ does not hold by assumption). But this contradicts Lemma 0.1. Hence B is not greater than A. □

Lemma 0.3. All subsets of any countable set are countable.

Proof. Recall that, by Exercise 7.1 of Chapter 3, every set of natural numbers is either finite or denumerable. Now consider any countable set A. Either it is finite or denumerable.

(1) On the one hand, suppose that A is finite. Then (by Exercise 6.1 of Chapter 3) all subsets of A are also finite.

(2) On the other hand, suppose that A is denumerable. Then it can be enumerated in some sequence $a_0, a_1, ..., a_n, ...$ Now consider any subset S of A. Then S can be put into a 1-1 correspondence with the set C of all n such that $a_n \in S$. But C is a set of natural numbers, meaning it is countable. Thus S is also countable. □

Lemma 0.4. The set of all sets of natural numbers with n as its greatest element is finite, in particular, it has 2^n elements.

Proof. By induction. Obviously the base case holds (as there is only one set of natural numbers with 0 as its greatest element). Suppose the result holds for some n. Now, the set of all sets of natural numbers with $n+1$ as its greatest element is just all sets of the form $\cup\{\{n+1\}, x\}$, where $x \in \mathcal{P}(n)$. But there are 2^{n+1} elemetns in $\mathcal{P}(n)$. Hence the set of all sets of natural numbers with $n+1$ as its greatest element has 2^{n+1} elements. □

Lemma 0.5. The union of a denumerable set of disjoint finite sets is denumerable.

Proof. Suppose we have a denumerable set S of disjoint finite sets $F_0, F_1, ..., F_n, ...$ For each F_i, let $N(F_i)$ be the size of F_i. Then define f_0 to be a function which assigns the elements of F_0 to the first $N(F_i)$ numbers. Then, for each n, let f_{n^+} be a function containing the ordered pairs of f_n, and if f_n contains a finite number of elements, add to f_{n^+} the ordered pairs which assigns the elements of F_{n^+} to the first $N(F_{n^+})$ elements of $\omega - \text{card}(f_n)$. If f_n is infinite, then take $f_{n^+} = f_n$. An obvious induction argument shows that each f_n must be finite, meaning another easy induction argument shows that each f_n has all elements of each F_i for $i \leq n$ in its domain. Then take the function f to be the set of all ordered pairs $\langle x, f_n(x)\rangle$ where $x \in F_n$. An easy induction argument shows that f is a 1-1 function between $\cup S$ and ω. □

Exercise 3.1

(1) Consider any denumerable set A. It can be placed in some sequence $a_0, a_1, ..., a_n, ...$ Then A can be put into a 1-1 correspondence with $A - \{a_0\}$, a proper subset of A.

(2) Consider some finite set D of denumerable sets. It can be placed in some sequence $a_0, ..., a_n$. For each a_i, let a_i' be the set of all elements x of a_i such i is the least index of D that contains x (that is, if $x \in a_j$ and $j \neq i$, then $j > i$). Obviously $\cup S = \cup\{a_0', ..., a_n'\}$, and all elements a_i' are disjoint with all other elements a_j', where $i \neq j$. Each a_i is denumerable, and each $a_i' \subseteq a_i$, meaning (by Lemma 0.3) that each a_i' is countable. Thus each a_i' is either finite or denumerable. Then the union of the set F of all finite a_i''s is finite. Then there are only a finite number of possible a_i''s that are denumerable. Say the set S of remaining a_i''s can be enumerated by the sequence $b_0, ..., b_m$, and, for each b_i, b_i can be enumerated as $c_0^i, c_1^i, ..., c_n^i, ...$ Then, for each k, the set S_k of all c_k^i (for all $i \leq m$) for every k is finite. Also, $\cup S = \{S_0, S_1, ..., S_n, ...\}$, which is denumerable by Lemma 0.5. Thus $\cup D = (\cup F) \cup (\cup S)$, which is denumerable by part (5) of this exercise.

(3) Suppose that A and B are denumerable. Then A can be placed in some sequence $a_0, a_1, ..., a_n, ...$ and B can be placed in some sequence $b_0, b_1, ..., b_n, ...$ Then, for each i, the set A_i of all ordered pairs $\langle a_i, b_j \rangle$ of $A \times B$ with $i \geq j$ is finite. Similarly, for each i, the set B_i of all ordered pairs $\langle a_j, b_i \rangle$ of $A \times B$ with $i \geq j$ is finite. For all i and j where $i \neq j$, $A_i \cup B_i$ is disjoint from $A_j \cup B_j$. Then $A \times B = \cup\{A_0 \cup B_0, A_1 \cup B_1, ..., A_n \cup B_n, ...\}$, which is denumerable by Lemma 0.5.

(4) Suppose that A is denumerable. Then it can be placed in some sequence $a_0, a_1, ..., a_n, ...$ Then the set of all finite subsets of A with i as its greatest index is finite, in particular, it has 2^i elements (the proof of this is identical to the proof of Lemma 0.4). For each i, let $S(a_i)$ be the set containing all subsets of A with a_i as its greatest element. Then the set of all finite subsets of A is just the set $\cup\{\{\emptyset\}, S(a_0), S(a_1), ..., S(a_n), ...\}$ which is denumerable by Lemma 0.4.

(5) Suppose that A is denumerable and F is finite. Say A can be enumerated in the sequence $a_0, a_1, ..., a_n, ...$ and F can be enumerated in the sequence $b_0, ..., b_m$. Then, if $F - A$ has k elements (it must be finite), we pair of the first k numbers with the elements of $F - A$, and then the rest of the numbers can be placed in a 1-1 correspondence with A itself.

Section 4

Exercise 4.1

Suppose that A is infinite and there is a choice function C for A. For any proper subset x of A, let $g(x) = x \cup \{C(A - x)\}$, and let $g(A) = A$. By Theorem 8.1 of Chapter 3, there is a function h from ω into $\mathcal{P}(A)$ such that $h(0) = g(0)$, and for any n, $h(n^+) = g(h(n))$. Then take a_0 to be the sole element of of $h(0)$ and for each n, take a_{n^+} to be the sole element of $h(n^+) - h(n)$. Now, a_0 is the sole element of $h(0) = g(0) = 0 \cup \{C(A - 0)\} = 0 \cup \{C(A)\} = \{C(A)\}$, that is, a_0 is the C-chosen element of A. Also note that $h(n^+) = g(h(n)) = h(n) \cup \{C(A - h(n))\}$, that is, $h(n^+)$ is the set of all elements of $h(n)$, as well as the C-chosen element of $A - h(n)$. We aim to show that a_{n^+} is the C-chosen element of $A - \{a_0, ..., a_n\}$. First we show that $h(n) = \{a_0, ..., a_n\}$. We have already proven the base case. Now suppose the result holds for some n. Then $h(n^+) = h(n) \cup \{C(A - h(n))\} = \{a_0, ..., a_n, C(A - h(n))\}$. But this means $a_{n^+} = C(A - h(n))$, meaning $h(n^+) = \{a_0, ..., a_n, a_{n^+}\}$, therefore completing the induction. Then a_{n^+} is the sole element of $h(n^+) - h(n) = (h(n) \cup \{C(A - \{a_0, ..., a_n\})\}) - h(n) = \{C(A - \{a_0, ..., a_n)\}$, that is, a_{n^+} is the C-chosen element of $A - \{a_0, ..., a_n\}$. Thus a_0 is the C-chosen element of A and a_{n^+} is the C-chosen element of $A - \{a_0, ..., a_n\}$, meaning each a_i is unique, and thus the set of all a_i is a denumerable subset of A.

Note that the second proof of Theorem 4.2 provided in the book can be expanded in essentially the same was as this exercise.

Section 6

Theorem 6.1

Here is why Σ is closed under chain unions. For consider some nest $\mathcal{N}$ of elements of Σ. Then $\cup\mathcal{N}$ is a 1-1 function (by Proposition 1.4 of Chapter 6). Consider the domain D of $\cup\mathcal{N}$. We show that the range R of $\cup\mathcal{N}$ is $D \cup D'$ (which would mean that $\cup\mathcal{N}$ is a 1-1 function from some subset S of A onto $S \cup S'$, meaning $\cup\mathcal{N} \in \Sigma$). If $x \in R$, then, for some function $f \in \mathcal{N}$, $x \in \text{Ran}(f)$. But $\text{Ran}(f) = \text{Dom}(f) \cup \text{Dom}(f)'$. Also $\text{Dom}(f) \subseteq D$ and $\text{Dom}(f)' \subseteq D'$. Thus $\text{Dom}(f) \cup \text{Dom}(f)' \subseteq D \cup D'$, meaning $x \in D \cup D'$. On the other hand, if $x \in D \cup D'$, either $x \in D$ or $x \in D'$. If $x \in D$, then for some function $f \in \Sigma$, $x \in \text{Dom}(f)$. But f is 1-1 onto $\text{Dom}(f) \cup \text{Dom}(f)'$, meaning $x \in \text{Ran}(f)$, and thus $x \in R$. If $x \in D'$, then for some $f \in \mathcal{N}$ and some $y \in \text{Dom}(f)$, $x = \pi(y)$. But f is 1-1 onto $\text{Dom}(f) \cup \text{Dom}(f)'$, meaning $x \in \text{Ran}(f)$, and thus $x \in R$. Hence $R = D \cup D'$, meaning $\cup\mathcal{N} \in \Sigma$.

It is remarked that $(M \cup M') \times (M \cup M') = (M \times M) \cup (M \times M') \cup (M' \times M) \cup (M' \times M')$. This is by Exercise 7.1 of Chapter 2.

It is remarked that $\phi \cup \psi$ is 1-1 from $M \cup M'$ onto $(M \times M)$. Here is why this is. We already know ϕ is a 1-1 map from M onto $M \times M$, and ψ is a 1-1 map from M' onto N. Because $N = \cup\{M \times M', M' \times M, M' \times M'\}$, N and $M \times M$ are disjoint (because M and M' are disjoint). This is why $\phi \cup \psi$ is 1-1.

Theorem 6.4

Here is why Σ is closed under chain unions. For consider some chain C of elements of Σ. Then $\cup C$ is a function, in fact a 1-1 map (by Proposition 1.4 of Chapter 6). Let D be the domain of $\cup C$. We need to show that the range of $\cup C$ is $D \times D$ (which would mean that $\cup C \in \Sigma$). On the one hand, suppose that $x \in \text{Ran}(C)$. Then, for some $f \in C$, $x \in \text{Ran}(f)$. But $\text{Ran}(f) = \text{Dom}(f) \times \text{Dom}(f)$, meaning $x \in D \times D$. On the other hand, suppose that $x \in D \times D$. Then $x = \langle y, z \rangle$ for some $y, z \in D$. Then, for some functions f and g (both in C), $y \in \text{Dom}(f)$ and $z \in \text{Dom}(g)$. But, f and g are comparable (because C is a chain). Without loss of generality, suppose that $f \subseteq g$. Then y and z are both in the domain of g, and because g is 1-1 onto $\text{Dom}(g) \times \text{Dom}(g)$, $x \in \text{Ran}(g)$, and thus $x \in R$. Hence $D \times D = R$, meaning $\cup C \in \Sigma$.

Note why, if we assumed that $A - M$ were not smaller than M, we could conclude that $A - M$ would include a subset M' of the same size as M. This is because we are assuming the axiom of choice, and thus the trichotomy of cardinals holds, meaning either $A - M \preceq M$ or $M \preceq A - M$. Because $A - M \prec M$ doesn't hold, this means that $M \preceq A - M$ holds.

It should be noted that M must be infinite (or else we could not use Theorem 6.3). For M is the domain of some function ϕ that is 1-1 from M onto $M \times M$, which means M must be infinite.

It is concluded from $M \cong A$ (say ϕ places M into a 1-1 correspondence with A) and $M \cong M \times M$, that $A \cong A \times A$. Here is why this is. First, $M \times M \cong M \times A$. This is because the set of all ordered pairs of the form $\langle \langle x, y \rangle, \langle x, \phi(y) \rangle \rangle$ where $x \in M$ and $y \in M$ is a 1-1 correspondence between $M \times M$ and $M \times A$. In a similar fashion, $M \times A \cong A \times A$. Thus $M \times M \cong A \times A$.

Section 7

Notes

When we are talking about isomoprhims from an ordinal α onto a set S of ordinals (such as in Lemma 7.2 and Proposition 7.3), it seems that the well ordering of S must be the well ordering of On.

In part 2 of Proposition 7.4, it is concluded that $\alpha < d$. Here is why this is: We know that $S \cong \alpha, c \cong S$, and $c \prec d$. Also, by Proposition 7.3, $\alpha \leq d$. If $\alpha = d$, then $c \cong d$, contrary to the fact that $c \prec d$. Thus $\alpha < d$.

In the same proof as the preceeding paragraph, it was concluded that $c \leq d$. Here is why this is: Suppose $c > d$. Then, as we have already shown in part (2) of the proof, $c \succ d$, contrary to the assumption that $c \prec d$.

There is in fact an alternative method of showing $c < d$ (in the second paragraph of Proposition 7.4 (2)): Suppose that $c \geq d$. Recall that $\alpha < d$ (meaning $\alpha < c$) and $c \cong \alpha$. But c is a cardinal, meaning it is not the same size as any lesser ordinal, and we have $c \cong \alpha$, an ordinal less than c. Thus this is a contradiction, meaning it must be that $c < d$.

Exercise 7.1

(a) Suppose L is a lower section of a well ordered system $\langle A, \leq \rangle$ and ϕ is an isomoprhism (with respect to $\leq$) from L into A. Say that an element x of L moves backwards if $x > \phi(x)$. Suppose some $x \in L$ moved backwards. Because L is a lower section of A, this would mean that $\phi(x) \in L$. Then, because ϕ is an isomorphism, $\phi(x) > \phi(\phi(x))$. Thus $\phi(x)$ also moves backwards. But this means that there is no least element of L which moves backwards, contrary to the fact that L is well ordered (because it is a subclass of A, which is well ordered). Hence no element of x moves backwards, meaning for every $x \in L$, $x \leq \phi(x)$.

(b) Here is how the above result yields both Theorem 2.1 of Chapter 6, and Lemma 7.2 of this chapter. In the case of Lemma 7.2, α is a lower section of On (because every ordinal is a lower section of On), and On is well ordered, and ϕ is an isomorphism (with respect to $\leq$) from L into On, meaning, by the above result, that for all $x \in \alpha$, $x \leq \phi(x)$. Now, recall that Theorem 2.1 of Chapter 6 stated that, for an isomorphism ϕ from a well ordered system $\langle A, \leq \rangle$ into itself, no element a moves backwards under ϕ. This follows from part (a) of this exercise, as ϕ is assumed to be an isomorphism from the whole of A, which itself is a lower section of A.